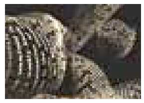
Codota AI
An Israeli startup that uses AI to allow developers to autofill code just raised $12 million in funding. **https://dgit.in/jul20-33**



Beauty in code
The genres of digital poetry, made from a multiple languages understood by humans and by computers. **https://dgit.in/jul20-34**

and analysis and not talk about Wireshark. The go-to choice for any forensic investigator when it comes to analysing network traffic, Wireshark is an immensely powerful tool and it was released in 1998. That was 22 years ago! Wireshark comes with numerous filters, giving you absolute control over the kind of data you want to capture. Two main categories of the filter being “capture filter” and “display filter”. Capture filter allows you to choose which network interface is needed to be monitored. It can be a WiFi connection, a VPN connection, a LAN network, raw USB traffic etc. Next is the display filters, which are applied over the captured packets to give finer control over the data. You can specify to show only traffic between specific IPs or traffic originating from specific ports. You can filter according to transport layer protocols (TCP/UDP) or application layer protocols (HTTP, FTP, ICMP, etc). Wireshark has the ability to follow a TCP stream to reveal all of the communication and it can even import all the objects involved in an HTTP communication, resulting in retrieval of images, important files etc. Wireshark is open-source and you can check it out here: https://dgit.in/wireshark.

- Tcpdump: Shockingly, this tool is a whopping ten years older than Wireshark and still is used widely even today! Think of tcpdump as the command line brother of Wireshark, but stronger. But there’s a catch. More intensive controls over data can result in quite complex commands. For example, let’s take this scenario. Each TCP packet has 6 flags, and each packet has this flag value set to 1 for specific kinds of packets. For example, if the TCP packet is an ACK (acknowledge) packet, quite understandably, the “ACK” packet is set to 1. Using tcpdump, you can filter the packet captures according to these flags. The complete commands and explanation of tcpdump are beyond the scope of this article, but for the curious minds, you should check this out: https://dgit.in/tcpdump. Tcpdump is available on most UNIX-like systems. There’s a Windows port called WinDump, which works on WinPcap (the Windows version of libpcap, a library dependency for tcpdump).

Image Forensics:

From figuring out whether an image is photoshopped or not, to deciphering complex hidden messages in the binary code of an image, this is where it is dealt with. Most tools here deal with steganography, the art of hiding data within seemingly commonplace stuff:

- Zsteg: A tool written in Perl, zsteg is used to detect and uncover data hidden using steganography in PNG/BMP files. The major techniques it can uncover are:
  - LSB steganography in PNG and BMP [https://dgit.in/LSB-steganography]
  - zlib-compressed data [https://dgit.in/Zlib]
  - OpenStego [https://dgit.in/openstego]
  - Camouflage 1.2.1[https://dgit.in/camouflage]
  - LSB with The Eratosthenes set [https://dgit.in/LSB]

  Zsteg can be installed as a ruby gem using the command `gem install zsteg`. A lot of different options are available, such as standalone or a combination, that can be used to uncover most of the hidden data.
- stegosolve: Working solely on .jpeg images, stegosolve is a tool written in Java with the ability to reveal data hidden inside images by changing the colour variations and frequencies. It performs analysis of the file structure and also allows each bit plane to be viewed on its own. Extracts from image bits are also taken into consideration and it comes with a packaged stereogram solver. Stegsolve is a single .jar file and can be executed on any platform with compatible java installation. You can follow these steps to get a functional stegosolve on your system (Note: only for unix systems):
  - `wget http://www.caesum.com/handbook/Stegsolve.jar -O stegsolve.jar`
  - `chmod +x stegsolve.jar`
- Then proceed with installation
- Online tools: There are some tools available online which perform forensic analysis of the images uploaded by the users and some may have facilities that other tools might not. Here are some of them:
  - FotoForensics
  - Forensically
  - Digital Image Forensic Analyzer

The tools we talked about were standalone, but no one takes a single handgun to a battlefield. Hence, there are multiple toolkits out there which are created for the sole purpose of DFIR and here are some of the best ones:

- SANS SIFT: Created by the SANS Institute, one of the monoliths of the security industry. The SANS Investigative Forensic Kit is considered the industry leader when it comes to DFIR-based distribution. Based on Ubuntu, SIFT is compatible with expert witness format, advance forensic format, raw format and memory analysis format. With auto-DFIR package update capabilities, SIFT provides cross-compatibility features through Windows and Linux.
- CAINE: A lesser known distribution, Computer Aided INvestigative Environment(CAINE), is still a tough contender in the list. Developed by Giovanni Bassetti in Italy, CAINE is a professional level DFIR distribution that was started in 2008. Again, like SIFT, CAINE is based on Ubuntu 16.04. CAINE 9.0 sets all the block